APPENDIX C

UNIVERSITI PENDIDIKAN SULTAN IDRIS

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SULTAN IDRIS EDUCATION UNIVERSITY

SOFTWARE DESIGN (SDD)

RESQ SISWA UPSI VOLUNTEER PORTAL

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REVISION HISTORY & DOCUMENT APPROVAL

Revision History

Date	Version	Author	Revision Details
25/9/2024	Version 1.0	SYASYA BINTI BASRI	First version created
10/10/2024	Version 2.0	SYASYA BINTI BASRI	Major changes in screen flow (wireframe) & description
23/11/2024	Version 3.0	SYASYA BINTI BASRI	Major change in section 3
29/12/2024	Version 4.0	SYASYA BINTI BASRI	Section 4 (Software Design) added

Document Approval

The following Software Design Document has been accepted and approved by the following :

SIGNATURE	NAME	DATE
Syasya	SYASYA BINTI BASRI	30/12/2024

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1.0 INTRODUCTION

This section provides an overview and scope description of the whole Software Design publication (SDD) publication. An explanation of this document's purpose is given, along with a list of acronyms and meanings.

1.1 Purpose of the application

The purpose of this Software Design Document (SDD) is to provide detailed information on the design, functions, and specifications of the ResQ Siswa UPSI Volunteer Portal. This document outlines the software and hardware components, as well as the detailed processes and data structures involved. It includes an architectural design, user interface design diagrams, and comprehensive descriptions of the design elements. The diagrams in this document adhere to the Unified Modeling Language (UML) standard. The primary goal of this document is to present the design flow of the ResQ Siswa UPSI Volunteer Portal, which aims to be a trusted platform for accessing volunteer events.

1.2 Scope

This document is written for theResQ Siswa UPSI Volunteer Portal. This document contains information about the system consideration, system architecture, low level design and high level design.

1.3 Definition, Acronyms, and Abbreviations

TERM	DEFINITION
ResQ Siswa UPSI Volunteer Portal	Name of the website
ResQ Siswa UPSI Volunteer Portal User	UPSI's Students and ResQ Siswa Club Organization

1.4 References

IEEE. IEEE Std 1016-1998 IEEE Recommended Practice for Software Design Descriptions. IEEE Computer Society, 1998.

1.5 Overview

The next two chapters of this document's parts cover the thorough design facets of the ResQ Siswa UPSI Volunteer Portal. ADV 1 (View 1), ADV 2 (View 2), and ADV 3 (View 3) are three different software architecture diagrams that are shown in Chapter 2. Each diagram is accompanied by detailed descriptions, enabling stakeholders to gain a deeper understanding of the application, its primary functionalities, and the corresponding architectural structure.

Chapter 3 encompasses the application's design goals, subsystems, and a decomposition table, which provides explicit definitions and clarifies the intended objectives of the ResQ Siswa UPSI Volunteer Portal. This section outlines the targeted goals of the application and its various subsystems, aiding stakeholders in comprehending the design intentions of theResQ Siswa UPSI Volunteer Portal thoroughly.

Furthermore, Chapter 4 describes the diverse interfaces associated with the ResQ Siswa UPSI Volunteer Portal. It encompasses a range of analytical diagrams and models that are precisely defined and explained. These diagrams and models serve the purpose of assisting stakeholders in understanding the underlying mechanisms and interactions within the application, ultimately contributing to a comprehensive grasp of the ResQ Siswa UPSI Volunteer Portal.

2.0 PROPOSED SOFTWARE ARCHITECTURE

2.1 View 1 (ADV 1)

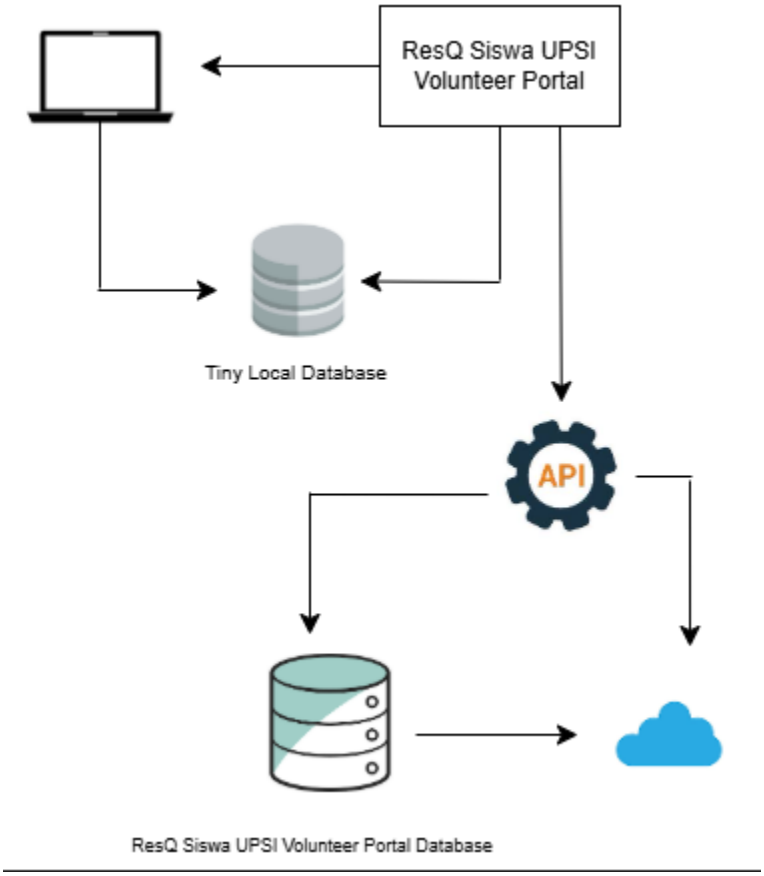

Diagram 1 : Architectural Design View (ADV 1)

2.2 View 2 (ADV 2)

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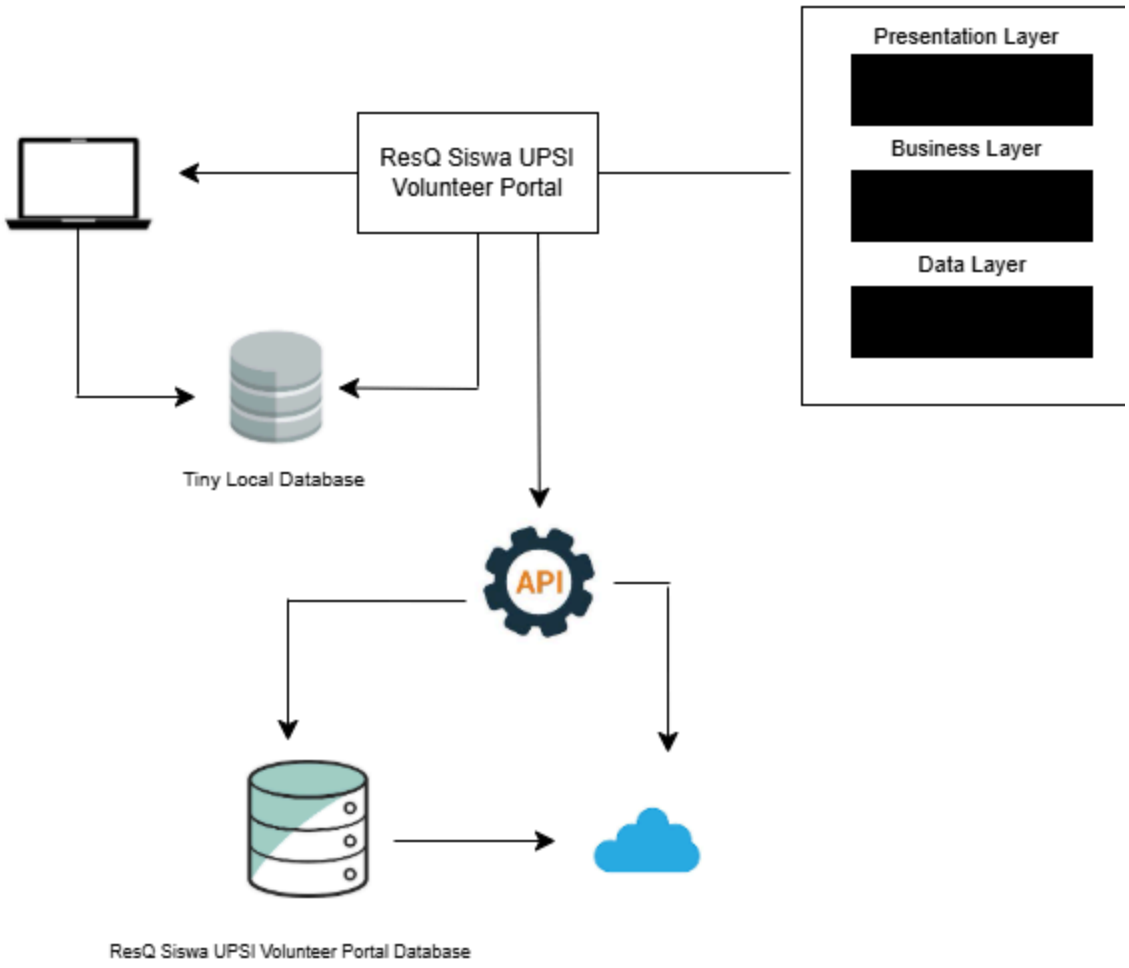


Diagram 2 : Architectural Design View (ADV 2)

2.3 View 3 (ADV3)

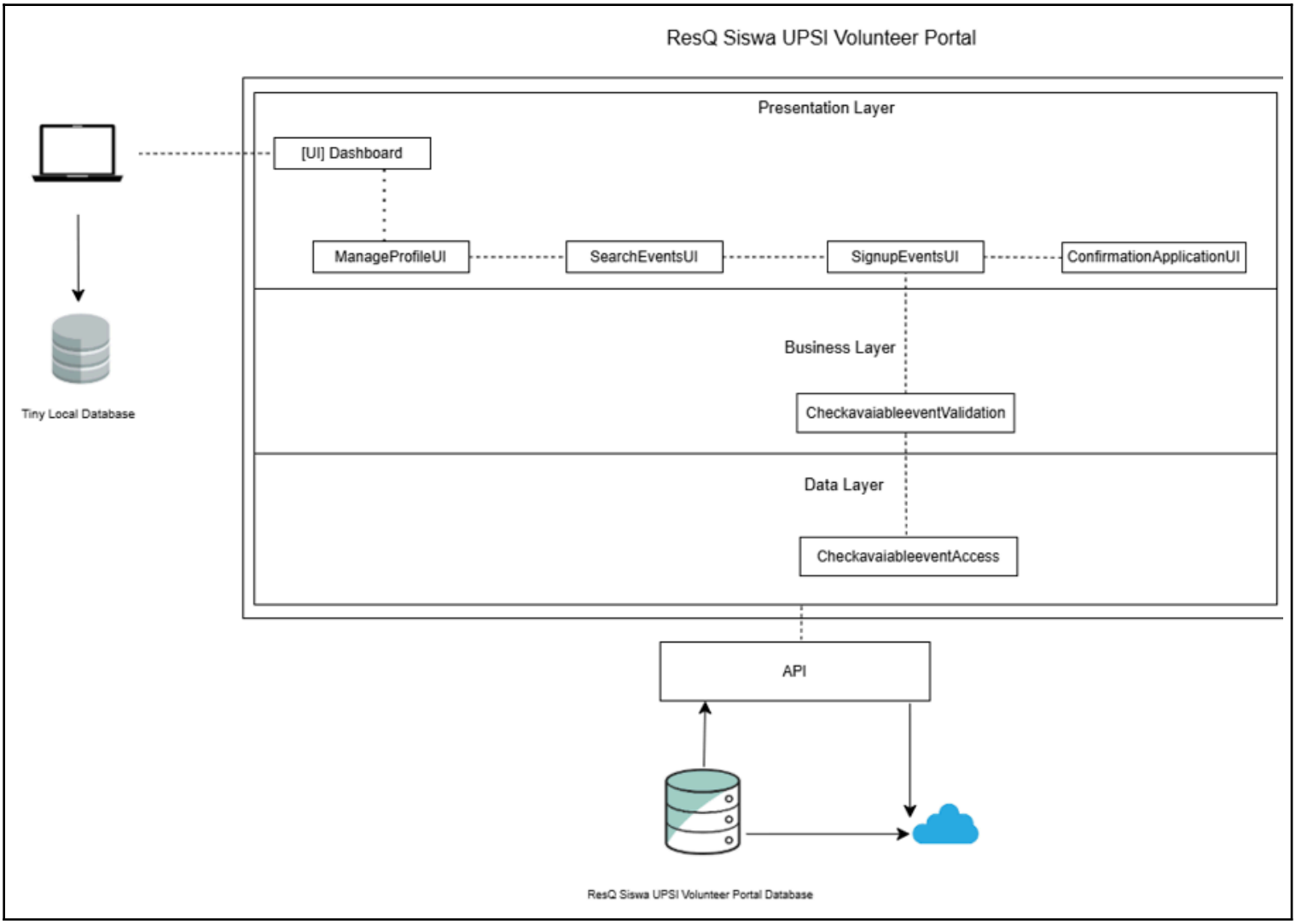


Diagram 3 : Architecture Design View (ADV3)

3.0 SUBSYSTEM DECOMPOSITION AND DESIGN GOALS

3.1 Subsystem Decomposition

3.1.1 Manage Profile (Student & organization)

Subsystem	Function	Presentation Layer	Business Layer	Data Layer
Manage Profile Subsystem	Manage Profile	Manage Profile UI Package **Component P1.1** UIDashboard **Component P1.2** UIEditprofile **Component P1.3** UIEditprofileSuccess **Component P1.4** UIEditprofileError	Manage Profile Validation Package **Component B1.1** getProfile() **Component B1.2** SendValidateData() **Component B1.3** ValidateWithDatabase() **Component B1.4** ProfileScreenDisplay() **Component B1.5** ProfileValidation()	Update Profile Data Access **Component D1.1** UpdateProdile() **Components D1.2** UpdateProfileData()

3.1.2 Search Events (Student)

Subsystem	Function	Presentation Layer	Business Layer	Data Layer
Search Events Subsystem	Search event availability	Search Events UI Package **Component P1.1** UIDashboard **Component P1.2** UIEventlist **Component P1.3** UIViewEventListSuccess **Component P1.4** UISearchEventListError	Ticket Price Validation Package **Component P2.1** getEvent List() **Component P2.2** SendEvent List() **Component P2.3** ValidateWithDatabase() **Component P2.4** Event ListScreenDisplay() **Component P2.5** EventListValidationResults()	Ticket Price Data Access **Component P3.1** EventLists_sendRequest() **ComponentP3.2** EventListDataAccess_receiveData()

3.1.3 View Events (Student)

Subsystem	Function	Presentation Layer	Business Layer	Data Layer
View Events Subsystem	View Available Events	View Events UI Package **Component P1.1** UIDashboard **Component P1.2** UIViewAvailableEvents **Component P1.3** UIAvailableEventSuccess **Component P1.4** UIAvailableEvent Error	Available Event Validation Package **Component P2.1** getAvailableEvent() **Component P2.2** sendValidateData() **Component P2.3** ValidatewithDatabase **Component P2.4** EventScreenDisplay() **Component P2.5** EventUpdateResult()	Available Event Access **Component P3,**1 UpdateAvailableEvent_SendRequest() **Component P3.2** UpdateAvailableevent_receiveData()

3.1.4 Sign up Event (Student)

Subsystem	Function	Presentation Layer	Business Layer	Data Layer
Sign up Event Subsystem	Sign up Event	Sign up Event UI Package **Component P4.1** UIDashboard **Component P4.2** UISignupEvent **Component P4.3** UISignupEventSuccess **Component P4.4** UISignupEventFail	Sign up Event Validation Package **Component B4.1** getEventDetail() **Component B4.2** sendValidateData() **Component B4.3** ValidateWithDatabase() **Component B4.4** decideScreenDisplay() **Component B4.5** receiveSignupDetailsResult()	Sign up Details Access **Component D4.1** UpdateEventDetail_sendRequest **Components D4.2** UpdateEventDetial_receiveData()

3.1.5 Confirmation Application (Student)

Subsystem	Function	Presentation Layer	Business Layer	Data Layer
Confirmation Application Subsystem	Confirmation Application	Confirm Application UI Subsystem **Component P5.1** UIDashboard **Component P5.2** ConfirmApplication_UI **Component P5.3** UIConfirmSuccess **Component P5.4** UIConfirmError	Confirmation Application Validation Package **Component B5.1** GetConfirmation() **Component B5.2** Confirmation() **Component B5.3** SendtoUpdateConfirmProgress() **Component B5.4** Update to DB **Component B5.5** ReceiveConfirmationResult() **Component B5.6** DisplayConfirmation()	ConfirmApplication Data Access **Component D5.1** TrackConfirmData_SendRequest() **Component D5.2** TrackConfirmData_ReceiveRequest()

3.1.6 Add events (Organization)

Subsystem	Function	Presentation Layer	Business Layer	Data Layer
Add Events Subsystem	View Events	**AddEventsUI Package** **Component P6.1** [UI]Dashboard **Component P6.2** AddEvent_UI **Component P6.3** UIAddSuccess **Component P6.4** UIAddError	**mapRender Package** **Component B6.1** getEventDetail() **Component B6.2** sendValidateData() **Component B6.3** ValidateWithDatabase() **Component B6.4** decideScreenDisplay() **Component B6.5** receiveSignupDetailsResult()	**dataAccess Package** **Component D6.1** UpdateEventDetail_sendRequest **Components D6.2** UpdateEventDetial_receiveData()

3.1.7 View Volunteer Profile (Organization)

Subsystem	Function	Presentation Layer	Business Layer	Data Layer
Add Events Subsystem	View Events	**AddEventsUI Package** **Component P6.1** [UI]Dashboard **Component P6.2** AddEvent_UI **Component P6.3** UIAddSuccess **Component P6.4** UIAddError	**mapRender Package** **Component B6.1** getEventDetail() **Component B6.2** sendValidateData() **Component B6.3** ValidateWithDatabase() **Component B6.4** decideScreenDisplay() **Component B6.5** receiveSignupDetailsResult()	**dataAccess Package** **Component D6.1** UpdateEventDetail_sendRequest **Components D6.2** UpdateEventDetial_receiveData()

3.1.8 Confirmation Student Application (Organization)

Subsystem	Function	Presentation Layer	Business Layer	Data Layer
Add Events Subsystem	View Events	**AddEventsUI Package** **Component P6.1** [UI]Dashboard **Component P6.2** AddEvent_UI **Component P6.3** UIAddSuccess **Component P6.4** UIAddError	**mapRender Package** **Component B6.1** getEventDetail() **Component B6.2** sendValidateData() **Component B6.3** ValidateWithDatabase() **Component B6.4** decideScreenDisplay() **Component B6.5** receiveSignupDetailsResult()	**dataAccess Package** **Component D6.1** UpdateEventDetail_sendRequest **Components D6.2** UpdateEventDetial_receiveData()

3.1.9 Manage User Account (Organization)

Subsystem	Function	Presentation Layer	Business Layer	Data Layer
Add Events Subsystem	View Events	**AddEventsUI Package** **Component P6.1** [UI]Dashboard **Component P6.2** AddEvent_UI **Component P6.3** UIAddSuccess **Component P6.4** UIAddError	**mapRender Package** **Component B6.1** getEventDetail() **Component B6.2** sendValidateData() **Component B6.3** ValidateWithDatabase() **Component B6.4** decideScreenDisplay() **Component B6.5** receiveSignupDetailsResult()	**dataAccess Package** **Component D6.1** UpdateEventDetail_sendRequest **Components D6.2** UpdateEventDetial_receiveData()

3.1.10 Access Analytics Data and Reports

Subsystem	Function	Presentation Layer	Business Layer	Data Layer
Add Events Subsystem	View Events	**AddEventsUI Package** **Component P6.1** [UI]Dashboard **Component P6.2** AddEvent_UI **Component P6.3** UIAddSuccess **Component P6.4** UIAddError	**mapRender Package** **Component B6.1** getEventDetail() **Component B6.2** sendValidateData() **Component B6.3** ValidateWithDatabase() **Component B6.4** decideScreenDisplay() **Component B6.5** receiveSignupDetailsResult()	**dataAccess Package** **Component D6.1** UpdateEventDetail_sendRequest **Components D6.2** UpdateEventDetial_receiveData()

3.2 Design Goals

DESIGN GOAL	QR_ID	QUALITY REQUIREMENTS	DESIGN CRITERIA GROUP	DESIGN CREATION	SUBSYSTEM INVOLVED
Design Goal 1	QR001	Event search will be display within 1 second	Performance	Response time	Search Event subsystem
	QR002	Available event will be show within 5 seconds	Performance	Response time	View Event subsystem
	QR003	A messages will be sent to student messages box after 24 hours if no action has been taken	Reliability	Availability	Confirmation subsystem
	QR004	If choose available event, it will direct to eventin less than 1 minute	Performance	Response time	Sign up event subsystem
	QR005	The event details constantly updated	Performance	Response time	Add events subsystem
Design Goals 2	QR006	The event details accurate	Reliability	Security	Add Event subsystem
	QR007	Data in ResQ Siswa UPSI Volunteer Portal is protected with password	Reliability	Security	Login subsystem

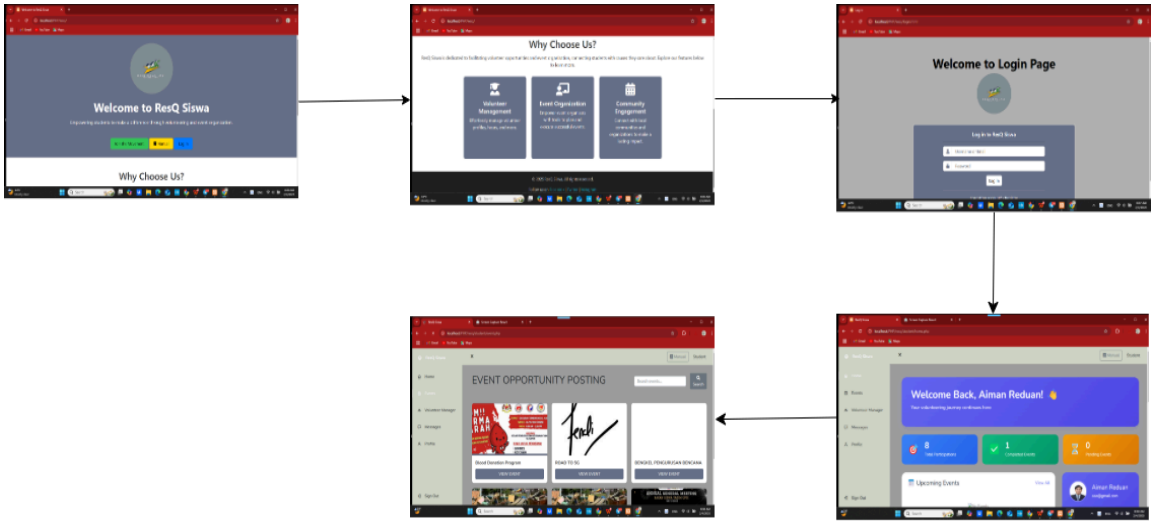
	QR008	Sign up event only can be made with authorization account	Reliability	Security	Sign up event subsystem
Design Goals 3	QR009	ResQ Siswa UPSI Volunteer Portal available 24/7	Reliability	Availability	-

4.0 DETAILED DESIGN

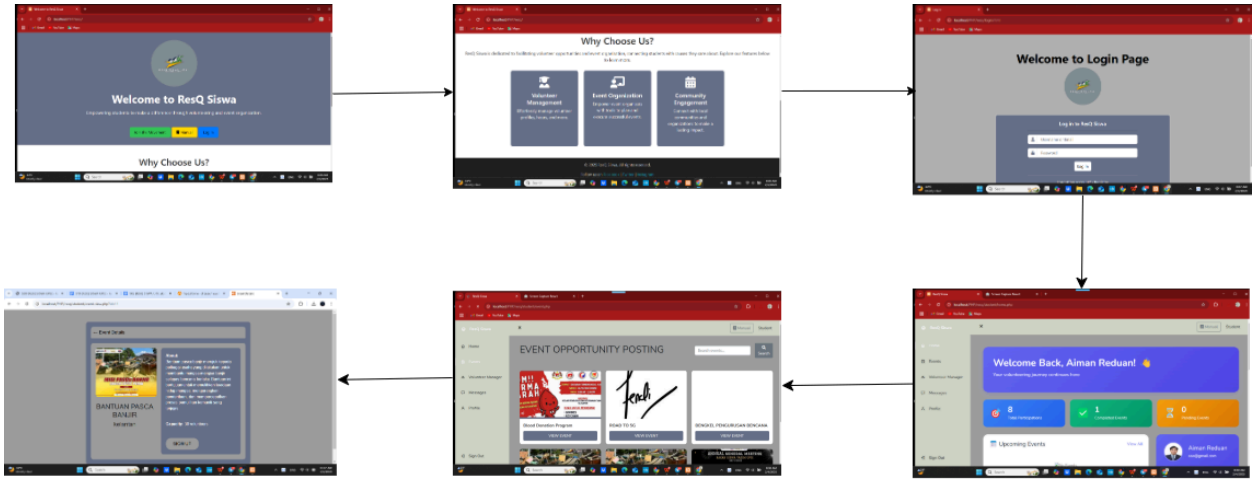
4.1 Navigation Design

4.1.1 Manage Profile

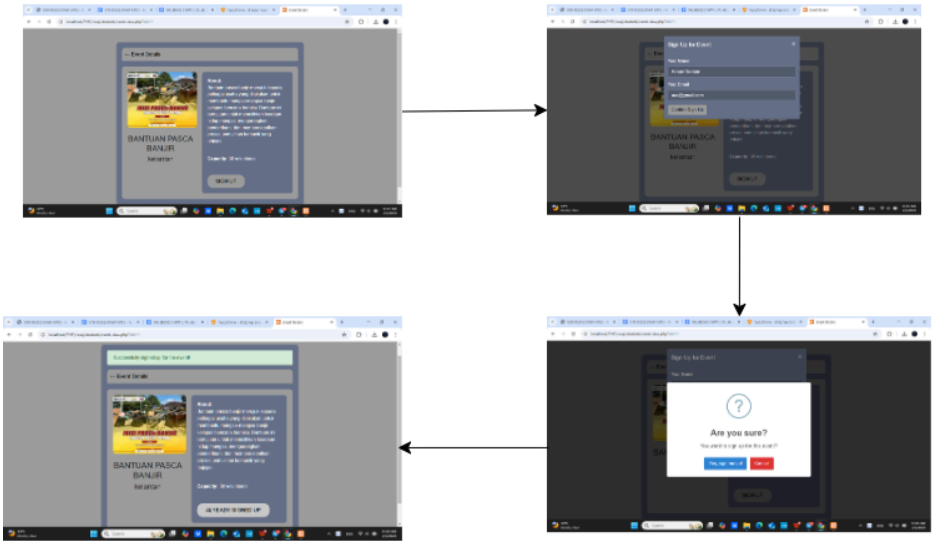
4.1.2 Search Event



4.1.3 View Event



4.1.4 Sign up Event



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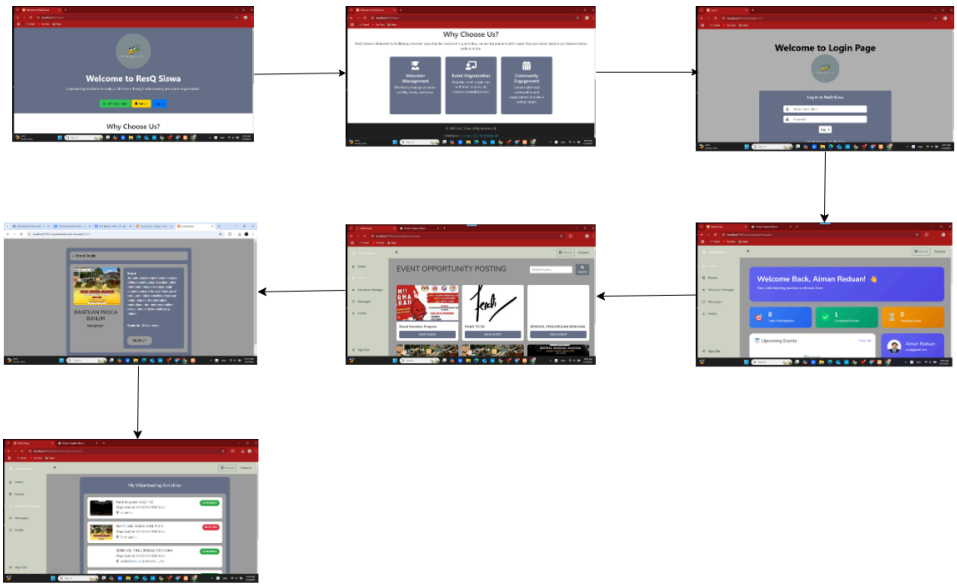
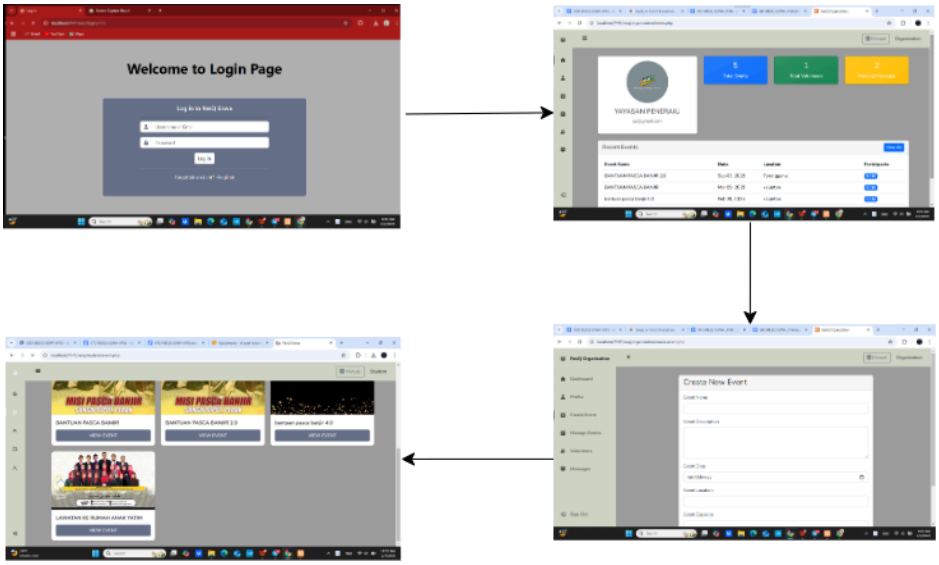
4.1.5 View Application

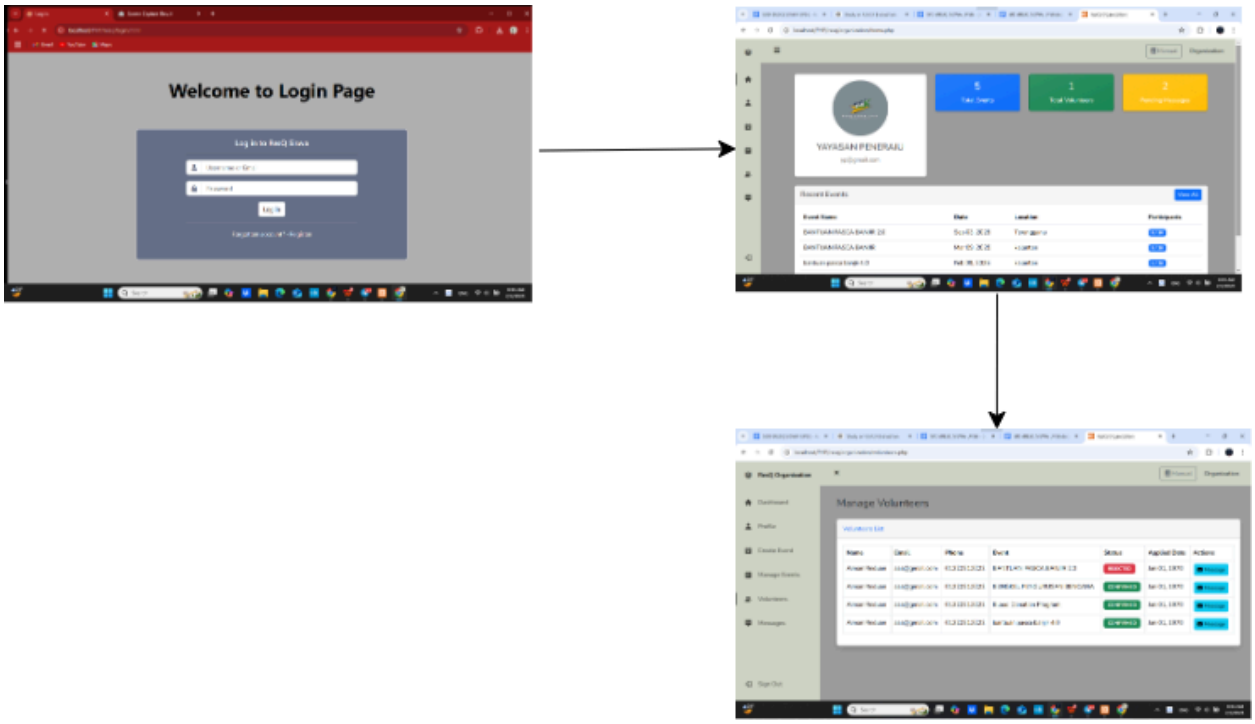


4.1.7 Confirmation Application

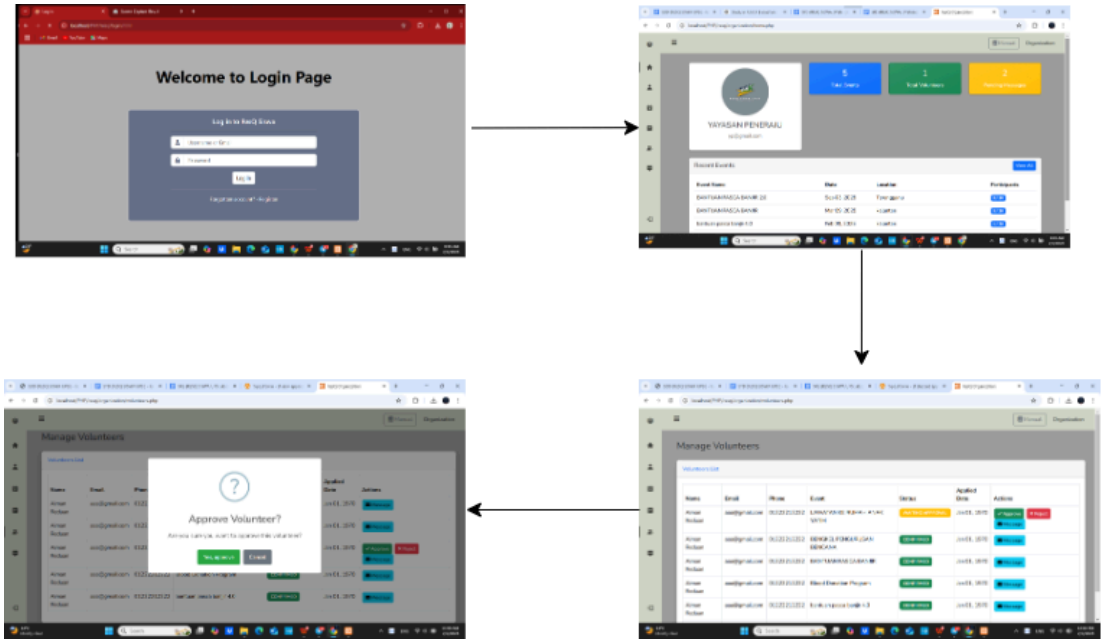
4.1.6 Add Events



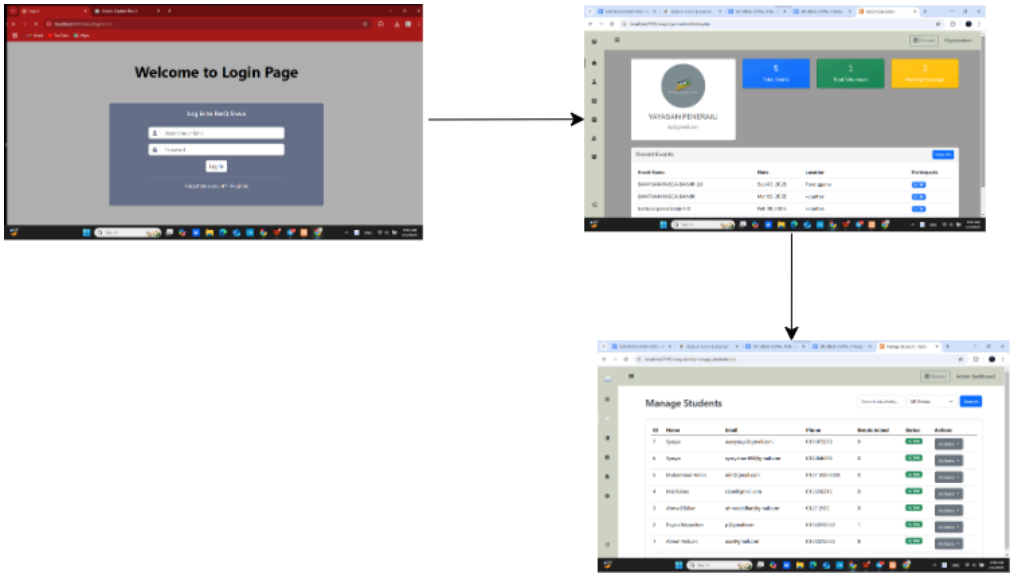
4.1.7 View Volunteer Profile



4.1.8 Confirm Student Application



4.1.9 Manage User Accounts



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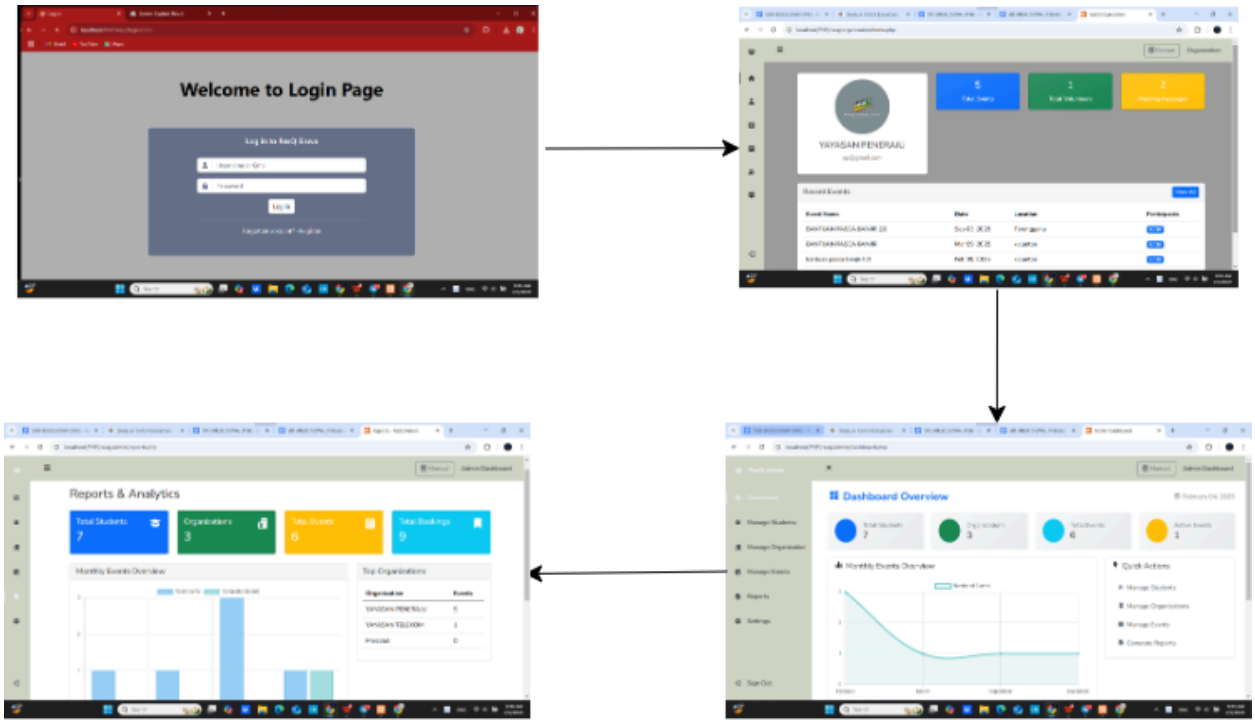
4.1.10 Access Analytics Data and Reports



4.2 Content Design

Content Table

Subsystem	Importance Level	Content Name	Content to be written in the screen
Manage Profile	High	Edit profile details	1) Fill in name 2) Fill inIC number 3) Fill in email 4) Fill in phone number
Search Event	High	Search Event	1) Name of the event available 2) Choose event
View Event	High	Show Event Details	1) Available Event Details
		Find Available Event	1) About event 2) Time, date and capacity volunteer of the event 3) Choose event
Sign up event	High	Choose available event	1) Choose event
Confirm Application	High	Show ApplicationDetails	1) Event details

			3) Button “Accept” to confirm and button “Reject” to not join the event.
Add event	High	Post Volunteer Opportunities	1) Add Event Details 2) Confirm add event
View Volunteer Profile	Medium	View Volunteer profile	1) User location 2) Station location details
		Discover volunteer	1) Choose Student based on their profile
Confirm Student Application	High	Confirmation Application	1) If accept, click accept button 2) If reject, click reject button
Manage User Account	High	Choose Account	1) List of student

		Remove account	1) Remove account if he student not activate the account
Access Analytics Data and Report	High	Click on report	1) View data and report that have been updated based on informations that have been edited